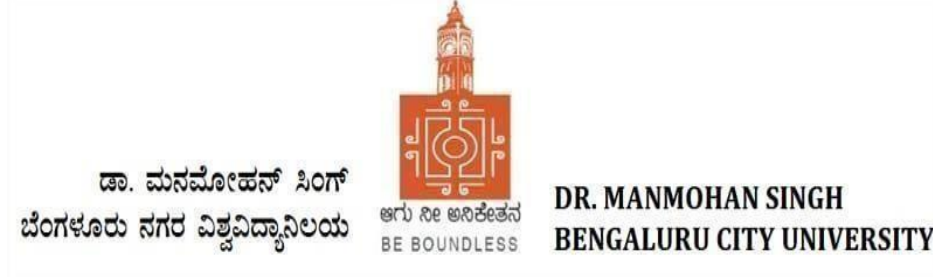


INTERNSHIP REPORT

On

“**DIGITAL PAYMENTS AND UPI ANALYSIS DASHBOARD**”

Submitted to Dr. Manmohan Singh Bengaluru City University



In partial fulfillment of the requirements for the award of the BCA degree

By

Chandra M S

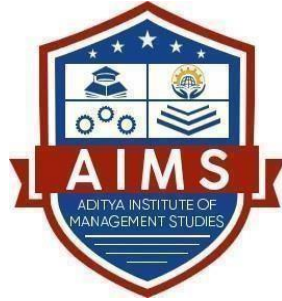
(U18AO23S0008)

under the guidance of

Mrs. Sangeetha R

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EVALUATION OF INTERNSHIP REPORT

For Submission to Dr. Manmohan Singh Bengaluru City
University

Student	Mentor/Supervisor
Name: Chandra M S	Name: Mrs. Sangeetha R
Reg. No.: U18AO23S0008	Signature:
Signature:	
Date:	Date:

Evaluation Summary

	Project Report	VIVA VOCE	Total
Maximum Marks	60	40	100
Marks Allotted			
Name of Examiner:	Signature:		

INTERNSHIP CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

MR. Chandra M S
INTERN ID: AF05112385

This is to certify that **Chandra M S**, a student of Bachelor of Computer Applications (BCA) from **Aditya Institute of Management Studies and Research**, Bangalore, has successfully completed a 3-month internship program at **Anudip Foundation**, from **12th February 2026** to **24th April 2026**.

During the internship, Candidate was actively involved in working on various projects related to Data Analysis and Business Intelligence, utilizing tools such as Power BI, Advanced Excel, and Python, MySQL. Candidate demonstrated a strong understanding of concepts, along with the ability to apply theoretical knowledge to practical business problems.

Throughout the internship period, candidate exhibited professionalism, dedication, teamwork, and a proactive attitude. Candidate shown ability to learn quickly, adapt to new challenges, and deliver quality work within deadlines was highly appreciated. Candidate also showed excellent communication and analytical skills while working on assignments and collaborating with peers.

Candidate's overall performance during the internship was found to be good.

We wish all the best for future academic and professional endeavors and we are confident that candidate will continue to achieve great success.

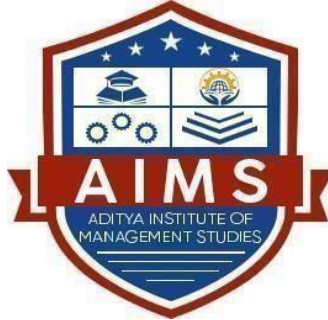


Upvan Dhar
Regional Manager

Anudip Foundation for Social Welfare

ADITYA INSTITUTE OF MANAGEMENT STUDIES

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COLLEGE CERTIFICATE

This is to certify that the Internship Report on “**Digital Payments and UPI Analysis Dashboard**” is carried out by **CHANDRA M S**, a student of VI Semester BCA, bearing **Register No. U18AO23S0008**. The student has successfully completed a four-week organizational study and has attended all the required guidance sessions.

The Internship Report is submitted to the Dr. Manmohan Singh Bengaluru City University in the Academic Year 2023-2026, in partial fulfilment of the requirements for the successful completion of VI semester BCA and the award of Bachelor of Computer Applications degree course of Dr. Manmohan Singh Bengaluru City University.

PLACE: BANGALORE

PRINCIPAL

DATE:

SUPERVISOR'S CERTIFICATE

This is to certify that the content of this report on “**Digital Payments and UPI Analysis Dashboard**” carried out by **CHANDRA M S**, a student of VI Semester BCA, Register No. U18AO23S0008, is a bonafide work carried out by the student under my supervision during the academic year 2023-26 in partial fulfillment of the requirements for the completion of VI Semester BCA Degree of Dr. Manmohan Singh Bengaluru City University.

This Internship Report is an original work carried out by the said student, and there is no plagiarism. This Internship report has not been submitted for the award of any other degree or any other institution.

PLACE: BANGALORE

Signature of the Supervisor

DATE:

DECLARATION OF STUDENT

I hereby declare that this Internship report on “Digital Payments and UPI Analysis Dashboard” is an original record of my experience while conducting the internship study at Anudip Foundation.

The organizational study was carried out by me in partial fulfillment of the requirements for the successful completion of the VI Semester BCA Course of Dr. Manmohan Singh Bengaluru City University. This Internship Report has not been submitted to any other university for the award of any degree.

PLACE: BANGALORE

Chandra M S

DATE:

U18AO23S0008

ACKNOWLEDGEMENT

I am indebted to our cherished Principal, **Prof. Dr. Pushpa Shetty**, Director of AIMS, whose support and permission were instrumental in undertaking this dissertation. Her guidance throughout this challenging endeavor has been invaluable.

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I am deeply indebted to **Mrs. Farheen Sultana Anjum** and **Ms. Ananya Nohria** from Anudip Foundation for providing me with the opportunity to carry out the organizational study and for their constant support, guidance, and encouragement throughout the internship.

Last but not least, I express my heartfelt gratitude to my beloved parents and dear friends for their unwavering encouragement and inspiration.

Chandra M S

U18AO23S0008

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1. INTRODUCTION TO INTERNSHIP

As part of my academic curriculum in Data Analytics and Visualization, I undertook an internship focused on the design and development of a Digital Payments and UPI Analysis Dashboard. This project-oriented internship provided me with valuable practical exposure to real-world financial data analysis and visualization techniques while enhancing my understanding of how data-driven insights can improve digital payment systems. The project also emphasized analyzing large-scale UPI transaction datasets to derive meaningful insights into transaction growth and user behavior.

The primary objective of this internship was to analyze Digital Payments and UPI transaction data and build interactive dashboards that effectively represent key performance indicators such as total transactions, transaction amount, success rate, monthly trends, and user behavior. The project aimed to help in understanding transaction patterns, evaluating payment performance, and supporting better decision-making. Additionally, key metrics such as Year-over-Year (YoY) growth, Month-over-Month (MoM) growth, and rolling mean trends were analyzed to provide deeper insights into transaction performance and consistency.

Throughout the internship, I was actively involved in the complete data analysis process. The project was developed using Power BI as the primary visualization tool. Data was collected, cleaned, and transformed to ensure accuracy and consistency. Various dashboards were created to present insights in a clear and interactive manner, including an Overview Dashboard, Trend Analysis Dashboard, Event/Category Analysis Page, and a detailed Drill-through Page for in-depth insights. These dashboards helped in understanding transaction flow, identifying trends, analyzing performance effectively, and evaluating how transaction volume and value evolved over time.

The internship experience helped me gain hands-on knowledge in data cleaning, data modeling, and creating calculated measures using DAX. I worked with different types of visualizations such as bar charts, line charts, pie charts, and KPI cards to effectively communicate insights. Additionally, I learned the importance of data accuracy, consistency, and proper visualization techniques in presenting meaningful results. The use of rolling mean analysis and trend indicators further enhanced the ability to identify long-term patterns and fluctuations in UPI transactions.

Along with technical skills, this internship also played a significant role in developing my soft skills, which are equally important in a professional environment. I improved my communication skills by clearly expressing ideas and explaining dashboard insights during discussions. I actively collaborated with peers, which enhanced my teamwork and coordination abilities. While working on the project, I strengthened my problem-solving skills by identifying data-related issues and resolving them effectively.

I also developed strong time management skills by completing tasks within deadlines and maintaining consistency throughout the internship. My presentation skills improved as I confidently presented my dashboards and explained key findings in a structured manner. Overall, this experience contributed to my professional growth by building confidence, adaptability, and a disciplined approach towards work.

This internship served as a bridge between theoretical knowledge and practical implementation. It enhanced my analytical thinking, problem-solving skills, and understanding of business intelligence concepts. By the end of the internship, I developed a deeper understanding of how data analytics can be used to analyze digital payment systems and improve financial decision-making. The project also

provided insights into how external factors such as seasonal trends and events can influence transaction behavior.

This internship has equipped me with essential skills in data analysis and visualization, preparing me for future roles in the field of data analytics and business intelligence. It also strengthened my ability to interpret financial data and convert it into actionable insights through interactive dashboards.

1.1 MOTIVATION

The motivation behind this project is to understand how data analytics can be used to improve digital payment systems, especially in the context of UPI transactions and financial technology. In today's competitive digital environment, businesses and users rely heavily on online payment methods for secure, fast, and convenient transactions. However, without proper analysis, it becomes difficult to measure transaction patterns, user behavior, and payment performance effectively.

This project was undertaken with the aim of gaining practical knowledge in analyzing digital payment data and converting it into meaningful insights. By working on real-world datasets, I was motivated to explore how different metrics such as total transactions, transaction value, payment success rate, and transaction trends influence the effectiveness of digital payment systems. Additionally, analyzing growth indicators such as Year-over-Year and Month-over-Month trends provided deeper insights into transaction performance and system efficiency.

Another key motivation was to develop strong skills in data visualization using tools like Power BI. Creating interactive dashboards helped in presenting complex data in a simple and understandable manner, enabling better decision-making. It also encouraged me to think analytically and identify patterns, trends, and variations in transaction data over time.

In addition to technical learning, this project also motivated me to improve my soft skills. I enhanced my communication skills by clearly explaining data insights and presenting dashboard results effectively. Working on the project helped me develop teamwork and collaboration skills through discussions and idea sharing. I also strengthened my problem-solving abilities by identifying data issues and resolving them systematically. Furthermore, I improved my time management skills by organizing tasks efficiently and meeting deadlines. My presentation skills were enhanced as I confidently explained my findings during reviews.

Overall, this experience contributed to my professional growth by building confidence, adaptability, and a structured approach towards work. It also strengthened my ability to apply analytical thinking in real-world scenarios and understand the importance of data-driven decision-making.

1.2 INTERNSHIP OBJECTIVE

The main objective of this internship is to gain practical knowledge in data analysis and visualization by working on a real-world project based on Digital Payments and UPI transaction analysis. The internship focuses on understanding how financial transaction data can be analyzed to evaluate payment trends, user behavior, and support better decision-making.

The specific objectives of this internship are:

- To collect, clean, and transform raw transaction data into a structured format suitable for analysis.
- To analyze key performance metrics such as total transactions, transaction value, payment success rate, and transaction trends.
- To design and develop interactive dashboards using Power BI for effective data visualization

- To use DAX formulas to create calculated measures and derive meaningful insights from transaction data.
- To understand user payment behavior and transaction performance through data analysis.
- To improve analytical thinking and problem-solving skills in a real-world scenario.

The internship aims to bridge the gap between theoretical knowledge and practical application by providing hands-on experience in business intelligence and data visualization. It also focuses on developing the ability to interpret financial data and convert it into actionable insights through interactive dashboards.

Through this internship, I gained exposure to handling large transactional datasets, performing UPI-based digital payment analysis, and deriving meaningful insights that help understand the growth, usage patterns, and future potential of digital payments in India.

1.3 SCOPE OF THE INTERNSHIP REPORT

The scope of this internship report focuses on analyzing and visualizing Digital Payments and UPI transaction data to understand payment behavior, transaction trends, and financial performance. The report covers the complete process of transforming raw transaction data into meaningful insights using data analytics and visualization techniques.

This internship involves working with datasets related to digital payments, including metrics such as total transactions, transaction value, payment success rate, failed transactions, and transaction frequency. The scope includes data cleaning, data transformation, and the creation of interactive dashboards using Power BI to present the analyzed results in a clear and effective manner. It also includes the use of

analytical measures such as growth trends and performance indicators to evaluate transaction behavior.

The report also highlights different aspects of transaction analysis, such as identifying trends, comparing transaction performance, and understanding user payment behavior. Various dashboards have been developed to represent key insights, including transaction overview, monthly payment trends, category analysis, and detailed transaction-level analysis.

In addition to technical aspects, the scope of this internship also includes the development of essential soft skills. Throughout the project, I enhanced my communication skills by effectively explaining data insights and participating in discussions. I improved my teamwork and collaboration abilities while working with peers and sharing ideas. The process of handling data challenges helped strengthen my problem-solving and analytical thinking skills.

Furthermore, I developed strong time management skills by organizing tasks efficiently and completing them within deadlines. My presentation skills improved as I confidently presented dashboards and explained key findings in a structured manner. Overall, the internship contributed to my professional growth by improving confidence, adaptability, and a disciplined approach to work.

However, the scope of this project is limited to data visualization and analysis only. It does not include the development of banking applications or live payment systems. The findings are based on the available dataset and are intended for learning and analytical purposes.

Overall, this internship report provides a comprehensive understanding of how data analytics can be applied in digital payment systems to support better decision-making and improve financial analysis.

2. INTRODUCTION TO COMPANY



Anudip Foundation is a non-profit organization established in 2007 with the aim of creating sustainable livelihood opportunities for underprivileged communities through skill development and digital training. Headquartered in Kolkata, India, the organization focuses on empowering youth and women from marginalized backgrounds by providing them with industry-relevant skills and employment opportunities.

The foundation was established by social entrepreneurs with a vision to bridge the gap between unemployment and skill-based education. Over the years, it has grown significantly and now operates multiple training centers across India, offering programs in information technology, digital skills, and professional development.

Anudip Foundation works closely with various corporate partners and organizations such as Accenture, Capgemini, Amazon, and Infosys to ensure that its training programs are aligned with current industry requirements. These collaborations help in providing placement opportunities and career support to learners after the completion of their training.

The organization has made a significant social impact by transforming the lives of more than 500,000 individuals through its training initiatives. It focuses on

equipping learners with both technical and soft skills, enabling them to secure sustainable employment and improve their standard of living.

Anudip Foundation's mission is to empower disadvantaged youth with market-driven skills and create digital livelihoods, while its vision is to promote inclusive growth and reduce inequality through education and employment opportunities.

Anudip Foundation plays a crucial role in bridging the digital divide and supporting economic development by preparing individuals for the modern workforce through technology-driven education and skill development programs.

2.1 KEY ASPECTS OF THE INTERNSHIP EXPERIENCE PROVIDED BY ANUDIP FOUNDATION

The internship at Anudip Foundation offered a comprehensive and practical learning experience by combining technical knowledge with real-world application. It provided exposure to industry practices and helped in understanding how data analytics is applied in real-time scenarios. The key aspects of the internship experience are as follows:

Exposure to Real World Projects:

I worked on a live project titled “Digital Payments and UPI Analysis Dashboard,” which helped me understand real-world transaction analysis processes, dashboard development, and data-driven decision-making in the financial domain. This experience enabled me to apply theoretical concepts to practical situations and gain deeper insights into financial data analysis.

Hands-on-Learning:

The internship provided practical exposure to industry-standard tools such as Power BI, Microsoft Excel, and DAX. I gained experience in data cleaning, transformation, modeling, and visualization processes, which enhanced my technical capabilities.

Additionally, I learned how to handle large datasets efficiently and create meaningful insights through interactive dashboards.

Collaboration with Industry Experts:

I received guidance from experienced mentors and professionals who provided valuable insights into real-world challenges, best practices, and industry expectations. This interaction helped me align my work with professional standards and improve the quality of my analysis and reporting.

Operational Understanding:

I gained knowledge about the structure and workflow of training programs, understanding how learning modules are designed to meet industry requirements and enhance skill development. This helped me understand how organizations structure their training processes to achieve better learning outcomes.

Student Interaction and Support:

I actively participated in discussions, collaborative sessions, and knowledge-sharing activities. This improved my communication skills and helped me effectively interact with peers and mentors while resolving queries related to the project. It also encouraged teamwork and active participation in group learning environments.

Innovation and Improvement:

I contributed to improving dashboard design and enhancing data visualization techniques to make transaction insights more clear, interactive, and user-friendly. This encouraged creativity, critical thinking, and the ability to present data in a visually effective manner.

Marketing and Analytical Exposure:

I developed an understanding of how data analytics supports digital payment systems. By analyzing UPI transaction data, I learned how transaction performance

can be evaluated and improved using data-driven insights. This also helped me understand how analytics contributes to strategic decision-making in financial systems.

Soft Skills Development:

In addition to technical learning, the internship significantly improved my soft skills. I enhanced my communication, teamwork, problem-solving, and time management abilities. I also developed confidence in presenting insights and working in a structured and professional manner.

These aspects collectively ensured a smooth and enriching learning experience, covering all stages from data handling and analysis to dashboard development and presentation. The internship at Anudip Foundation provided a strong foundation in both technical and professional skills, preparing me for real-world industry challenges.

3. INTRODUCTION TO THE PROJECT

The Digital Payments and UPI Analysis Dashboard project began with understanding the dataset, identifying key transaction metrics, and defining the objectives of the analysis. As part of the internship, I focused on applying my data analytics and visualization skills to design and develop an interactive and user-friendly dashboard. The project also involved understanding transaction patterns and analyzing payment behavior to generate meaningful business insights. This included preparing the data, planning the dashboard layout, and identifying important performance indicators for effective analysis.

Using tools like Power BI for visualization, I worked on developing dashboards that represent key metrics such as total transaction volume, total transaction value, year-over-Year (YoY) growth percentage, Month-over-Month (MoM) growth, and transaction trends over time. These metrics helped in evaluating overall performance and identifying patterns in digital payment adoption.

Special emphasis was placed on creating clear and interactive visualizations, including KPI cards, bar charts, line charts, donut charts, and comparison visuals, to ensure effective data representation and easy interpretation. These visual elements improved the usability of the dashboard and made complex data more understandable for users.

The project also included data cleaning, transformation, and the use of DAX formulas to create calculated measures such as rolling mean values, growth percentages, and transaction comparisons across different years and periods. Slicers and filters were implemented to allow dynamic interaction with the data, enabling users to explore insights based on different parameters such as year, month, transaction value, and seasonal trends. This interactive approach enhanced the

analytical capabilities of the dashboard.

The dashboard includes multiple analytical sections such as transaction overview, growth trend analysis, event-based comparison, and yearly insights. It helps in understanding how digital payment transactions have evolved over time and how different external factors such as festive seasons and COVID periods have influenced UPI transaction behavior.

The goal of this project was to analyze digital payment transaction data and provide meaningful insights that support a better understanding of transaction growth, financial trends, and user payment activity. The dashboard allows users to monitor transaction performance, compare yearly growth, evaluate monthly trends, and understand transaction distribution patterns, contributing to better financial analytics and data-driven decision-making.

3.1 ABOUT THE PROJECT

The Digital Payments and UPI Analysis Dashboard is a comprehensive data visualization project developed during my internship to analyze and understand digital payment transaction patterns. The project focuses on transforming raw transaction data into meaningful insights using business intelligence techniques. It helps in studying UPI transaction growth, payment trends, transaction values, and yearly performance through interactive dashboards, enabling better interpretation of financial data.

The project began with an in-depth analysis of the dataset and understanding the requirements of digital payment analytics. By studying transaction trends and financial metrics, I developed a structured plan to create dashboards that are interactive, user-friendly, and informative. The goal was to ensure clarity in data

representation and provide actionable insights for understanding transaction growth and payment activity.

The dashboard was developed using Power BI, which provides a powerful platform for creating interactive reports and visual analytics. Various visual elements such as KPI cards, bar charts, line charts, area charts, trend comparison visuals, and distribution charts were used to represent transaction data effectively. The interface was designed to be clean, responsive, and easy to navigate, allowing users to quickly understand key insights and analyze transaction performance across different time periods.

Data preparation was an important part of the project. The dataset was cleaned and transformed to remove inconsistencies and ensure data accuracy. Missing values, formatting issues, and duplicate records were handled carefully before analysis. DAX (Data Analysis Expressions) was used to create calculated measures such as Year-over-Year (YoY) Growth Percentage, Month-over-Month (MoM) Growth, Rolling Mean Trends, and transaction comparisons. These measures helped in deriving deeper insights from transaction data and identifying performance patterns over time.

The project includes several important dashboard modules:

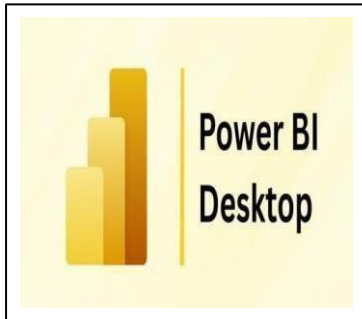
- **UPI and Digital Payments Overview Dashboard:** Provides a summary of important KPIs such as total transaction volume, total transaction value, YoY Growth Percentage, MoM Growth Percentage, and monthly growth indicators.
- **UPI Growth Trend Dashboard:** Displays transaction volume and transaction value trends across different years to understand overall growth in digital payments and analyze long-term transaction performance.

- **Monthly Transaction Analysis Dashboard:** Shows monthly transaction distribution and rolling mean trends to identify transaction patterns, seasonal changes, and monthly performance variations.
- **Event Impact Dashboard:** Focuses on analyzing the impact of COVID and festive seasons on UPI transaction volume and transaction value to understand how external events influence digital payment activity.
- **Transaction Insights Dashboard:** Provides yearly growth comparison, transaction value distribution, average transaction trends, and rolling mean analysis for deeper financial understanding.
- **Data Interaction Features:** Slicers and filters were implemented to allow dynamic analysis based on year, month, transaction value, and growth trends, improving user interaction and dashboard flexibility.

Throughout the project, emphasis was placed on data accuracy, visualization clarity, and user interaction. This project enhanced my skills in data analytics, dashboard design, and business intelligence, while improving my ability to interpret financial data and present insights effectively. Overall, the Digital Payments and UPI Analysis Dashboard demonstrates how data visualization simplifies complex transaction datasets and supports data-driven decision-making in digital payment analytics.

3.2 TECHNOLOGY USED

This project involved the use of modern data analytics and visualization tools to develop an interactive and efficient Digital Payments and UPI Analysis Dashboard. The focus was on creating a user-friendly, dynamic, and insightful dashboard that helps in analyzing transaction data effectively and supports better decision-making.



The primary tool used in this project is Microsoft Power BI, which was utilized for data visualization and dashboard creation. Power BI provides powerful features such as interactive charts, KPI cards, slicers, filters, and drill-down capabilities, enabling users to explore the transaction data easily and gain meaningful insights.



For data handling and preprocessing, Microsoft Excel was used extensively in this project. It played a key role in managing the raw dataset related to Digital Payments and UPI transactions. Excel was used to organize the data into a structured format, perform initial data cleaning by removing duplicates and handling missing values, and correct formatting issues to ensure consistency. Basic operations such as sorting, filtering, and simple calculations were also carried out to prepare the dataset before importing it into Power BI.



DAX (Data Analysis Expressions) was used within Power BI to create calculated measures and perform advanced data analysis. Important metrics such as Year-over-Year (YoY) Growth Percentage, Month-over-Month (MoM) Growth Percentage, Rolling Mean Trends, Total Transaction Volume, and Transaction Value comparisons were calculated using DAX formulas. These measures helped in identifying trends, patterns, and performance variations over time.

The project also involved data modeling techniques, where relationships between different data fields were established to ensure accurate analysis and reporting. Proper structuring of the dataset improved dashboard performance, accuracy, and usability. Additionally, various visualization techniques were used, including bar charts, line charts, area charts, KPI cards, distribution charts, and comparison visuals to represent transaction data clearly and effectively.

Overall, the technologies used in this project ensured efficient data processing, accurate analysis, and interactive visualization, making it easier to understand UPI transaction trends and support data-driven decision-making.

3.3 FRONTEND DEVELOPMENT

To develop an interactive and user-friendly interface for data visualization, I used the following tools and techniques:

- **Power BI**

Microsoft Power BI is a powerful data visualization tool developed by Microsoft, ideal for creating interactive dashboards and reports. It allows efficient representation and real-time updating of data through dynamic visuals. In this project, Power BI was used to create multiple dashboards such as transaction overview, growth trend analysis, monthly transaction analysis, event impact analysis, and transaction insights, ensuring a smooth and interactive user experience.

- **Data Visualization Techniques**

Data visualization techniques include the use of charts such as bar charts, line charts, area charts, pie charts, donut charts, KPI cards, and distribution charts. These visuals help in presenting complex transaction data in a simple and understandable format. They allow users to easily identify trends, compare transaction performance, and gain meaningful insights from financial data.

- **DAX (Data Analysis Expressions)**

DAX (Data Analysis Expressions) is used to create calculated measures and perform advanced data analysis. It helps in deriving important metrics such as Year-over-Year (YoY) Growth Percentage, Month-over-Month (MoM) Growth Percentage, Rolling Mean Trends, and Transaction Value comparisons, which are essential for evaluating UPI transaction performance.

Additionally, interactive features such as slicers and filters were implemented to allow users to explore the data dynamically based on different parameters such as year, month, transaction value, and growth trends. These features enhanced user interaction and made the dashboard more flexible and insightful.

3.4 BACKEND (DATA PROCESSING & MODELING)

The backend of the Digital Payments and UPI Analysis Dashboard focuses on data processing, transformation, and modeling to ensure accurate and meaningful analysis. Unlike traditional application development, the backend in this project is handled within Microsoft Power BI and Microsoft Excel, where data is processed and prepared for visualization.

The first step involved importing the transaction dataset into Power BI. Before visualization, the data was cleaned and transformed using Power Query Editor. This process included removing duplicate values, handling missing values, correcting data types, and organizing the dataset into a structured format suitable for analysis. These steps ensured data quality and reliability for further processing.

Data modeling was performed to establish relationships between different data fields such as year, month, transaction volume, transaction value, growth percentages, festive indicators, and COVID period classification. Proper relationships ensured accurate calculations, reduced data redundancy, and enabled efficient data flow

across different dashboard modules.

DAX (Data Analysis Expressions) was used extensively to create calculated measures and perform advanced analysis. Important metrics such as Total Transaction Volume, Total Transaction Value, Year-over-Year (YoY) Growth Percentage, Month-over-Month (MoM) Growth Percentage, Rolling Mean Trends, and transaction comparisons were calculated using DAX formulas. These measures played a key role in analyzing trends and performance over time.

The backend also ensures that all visualizations are dynamically connected to the data. Any changes made through slicers or filters are processed instantly, providing updated and real-time insights to the user. This improves the responsiveness and interactivity of the dashboard.

Power Query was used to transform raw transaction data into a structured format, improving overall consistency and optimizing it for better dashboard performance and reporting accuracy. Overall, the backend converts raw data into meaningful information, enabling accurate analysis and supporting interactive dashboards.

Overall, the backend converts raw UPI transaction data into meaningful information, enabling accurate analysis and supporting interactive dashboards. It forms the foundation of the project by ensuring that data is clean, structured, and effectively organized for visualization and business intelligence reporting.

3.5 DATABASE

The database used in this project consists of structured digital payment and UPI transaction data collected for analysis and visualization purposes. The dataset serves as the foundation for the Digital Payments and UPI Analysis Dashboard and contains transaction-related information required for understanding payment trends, transaction growth, and financial performance across different periods.

The dataset was initially prepared using Microsoft Excel and later imported into Power BI for analysis and dashboard development. Proper organization of the database ensured smooth processing, accurate calculations, and reliable visualization outputs. It also helped in maintaining data consistency across different dashboard modules.

The database includes important transaction-related fields such as year, month name, month number, transaction volume, transaction value, Year-over-Year (YoY) growth percentage, Month-over-Month (MoM) growth percentage, rolling mean values, financial year classification, festive season indicators, and COVID period indicators. These fields support detailed transaction analysis and help in identifying patterns and trends in digital payment behaviour.

The database was cleaned and structured to remove duplicate entries, handle missing values, and correct formatting inconsistencies. Proper data preparation improved the quality of analysis and ensured that dashboard results remained accurate, consistent, and reliable.

A well-maintained database plays an important role in dashboard development because it supports efficient analysis, faster data processing, and meaningful insight generation. It also ensures consistency between calculations, visualizations, and reporting outputs. Overall, the database component of this project ensured that data was well-organized, accurate, and ready for analysis, forming a strong foundation for creating meaningful insights and effective visualizations.

4. PROJECT CODE

The Digital Payments and UPI Analysis Dashboard is developed using Power BI, where the project logic is implemented using DAX (Data Analysis Expressions). Instead of traditional programming code, this project uses calculated measures, data transformation techniques, and data modeling to analyze transaction data effectively. DAX formulas help generate KPIs, identify transaction growth, and create meaningful insights from the dataset.

➤ KPIs & DEFINITIONS

KPI	Definition	Formula	Example Use
Transaction Volume	Total number of UPI transactions processed	Sum of Transaction Volume	Measures transaction activity
Transaction Value	Total monetary value of UPI transactions	Sum of Transaction Value	Measures financial performance
YoY Growth %	Percentage growth compared to previous year	$((\text{Current Year} - \text{Previous Year}) \div \text{Previous Year}) \times 100$	Analyze yearly growth
MoM Growth %	Percentage growth compared to previous month	$((\text{Current Month} - \text{Previous Month}) \div \text{Previous Month}) \times 100$	Analyze monthly trends
Average Transaction Value	Average value per transaction	$\text{Transaction Value} \div \text{Transaction Volume}$	Measure spending pattern
Rolling Mean Trend	Smoothed average trend across periods	Rolling average calculation	Identify long-term trends
COVID Period Analysis	Compares COVID vs Non-COVID transactions	Based on COVID indicator	Impact analysis
Festive Season Analysis	Compares festive vs non- festive transaction trends	Based on festive indicator	Seasonal analysis
Monthly Growth Indicator	Tracks increase/decrease in monthly transactions	Monthly growth calculation	Performance tracking
Yearly Trend Analysis	Shows long-term yearly transaction trend	Year-wise comparison	Growth analysis

➤ KPI METRICS

- **Transaction Volume: 676.49K:** Represents the total UPI transaction volume recorded across the dataset, indicating strong digital payment activity.
- **Transaction Value: 99.99M:** Represents the total monetary value of transactions processed through UPI platforms.
- **YoY Growth %: 60.59K:** Indicates yearly growth in transaction volume, showing increasing adoption of digital payments over time.
- **MoM Growth %: 2.23K:** Represents monthly growth in transaction activity, helping identify short-term performance changes.
- **Rolling Mean Volume: 655.02K:** Used to smooth transaction fluctuations and identify long-term transaction patterns.
- **Average Transaction Value: 99.99M:** Indicates the average value per transaction, helping analyze user spending patterns.
- **Growth Indicator: 11.04K:** Shows transaction growth progression and overall performance movement across different periods.

❖ KEY INSIGHTS

- Transaction volume and value have increased consistently across years, indicating growing reliance on UPI-based payments.
- Higher transaction volume and value reflect increased digital payment adoption and stronger financial activity.
- YoY and MoM growth metrics help measure transaction consistency and market expansion.
- Rolling mean analysis helps identify long-term payment behavior and transaction stability.

➤ TRANSACTION TREND ANALYSIS

- **Monthly Transaction Trend (Line Chart)**

- o Displays transaction growth across months.
- o Helps identify seasonal transaction behavior.

Insight: Higher transaction activity is visible during certain months, indicating increased payment demand and seasonal patterns.

- **Yearly Transaction Trend (Bar/Area Chart)**

- o Shows year-wise increase in transaction volume and transaction value.
- o Helps understand long-term UPI growth.

Insight: Continuous yearly growth indicates increasing adoption of digital payments.

➤ EVENT-BASED ANALYSIS

- **COVID vs Non-COVID Analysis**

- o Compares transaction performance during pandemic and non-pandemic periods.
- o Helps identify behavioral changes in digital payments.

Insight: UPI usage increased during COVID due to contactless payments.

- **Festive vs Non-Festive Analysis**

- o Measures transaction activity during festive periods.
- o Highlights seasonal transaction spikes.

Insight: Festive seasons contribute to higher transaction volume and value.

➤ DASHBOARD INTERACTION

- Filters and slicers are used to dynamically analyze data by year, month, and transaction trends.
- Interactive visuals help users explore transaction insights more effectively.
- KPI cards provide quick understanding of transaction performance.

Overall, the project code and DAX calculations helped transform raw transaction data into meaningful insights, supporting effective dashboard development and accurate financial analysis.

Year	Transaction Volume	Transaction Value	YoY Growth %	MoM Growth %	Rolling Mean Volume
2020	0.02M	1.88M	—	—	0.01M
2021	0.04M	7.32M	Growth Started	Moderate	0.03M
2022	0.08M	12.79M	Increased	Stable	0.05M
2023	0.12M	18.32M	Strong Growth	Positive	0.06M
2024	0.18M	25.36M	High Growth	Stable	0.07M
2025	0.23M	30.53M	Peak Growth	Positive	0.07M

Insight: Transaction volume and value increased consistently over the years, Showing strong adoption of UPI payments. Recent years indicate higher growth due to increased digital payment acceptance and improved transaction accessibility.

4.1 OUTPUT

Tren...

Eve...

Ins...

UPI & Digital Payments Overview Dashboard

Years

Select...

2016

2017

2018

2019

2020

2021

2022

2023

UPI Digital Transaction Volume Trend

Sum of Volume_Mn by Year



Monthly UPI Transaction Volume

Sum of Volume_Mn by Month_Name



UPI Growth Trend: Volume vs Value

Sum of Volume_Mn and Sum of Value_Cr by Year



UPI Volume Trend (Rolling Mean)

Sum of Volume_RollMean_3M by Month_Name



Total Transaction Volume
676.49K

YoY Growth %
60.59K

MoM Growth %
2.23K

Total Transaction Value
99.99M

Over...

Events

Insig...

UPI Transaction Trend Analysis

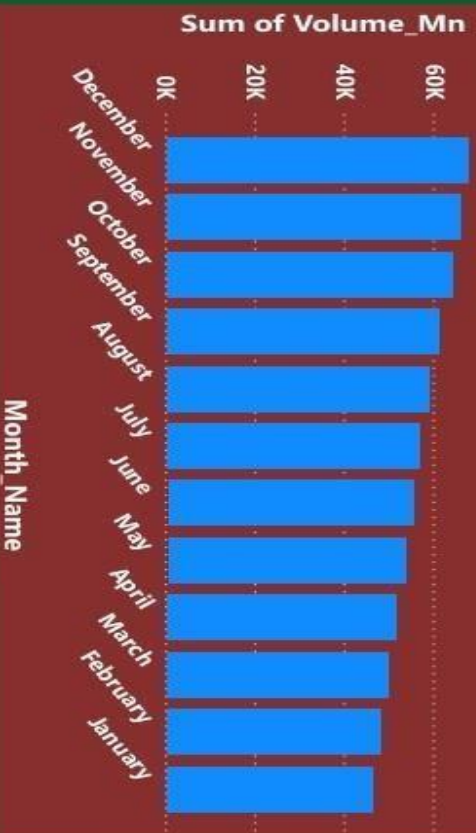
pure growth trend over time

Sum of Volume_Mn by Year



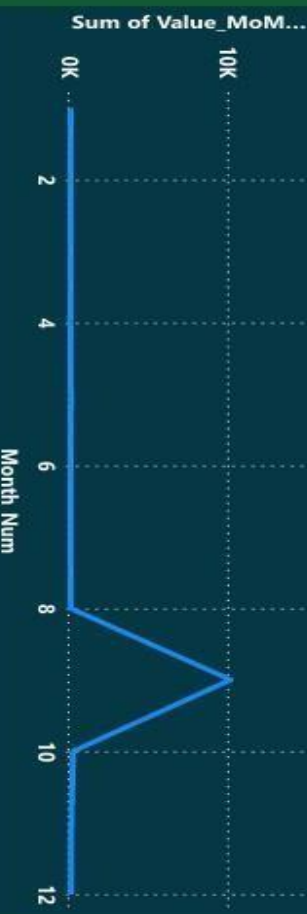
Monthly UPI Volume Distribution

Sum of Volume_Mn by Month_Name



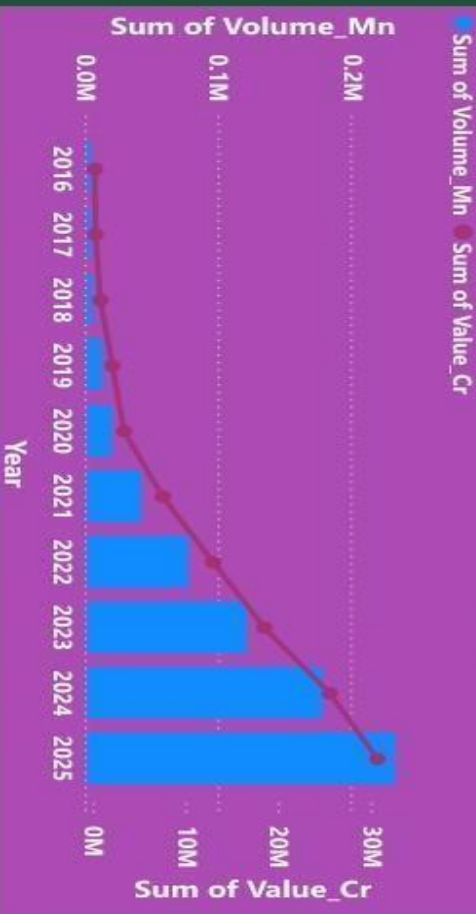
Monthly Value Growth Trend (MoM %)

Sum of Value_MoM_% by Month_Num



UPI Transaction Trend: Volume vs Value

Sum of Volume_Mn and Sum of Value_Cr by Year



Over...

Trends

Insig...

Event Impact on UPI Transactions

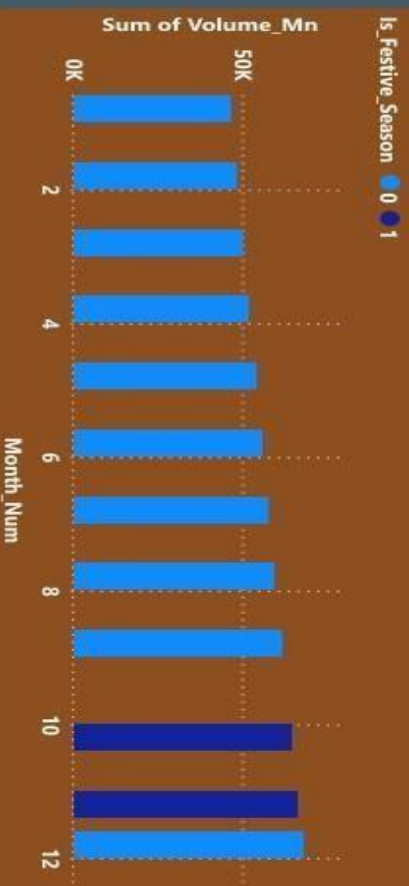
UPI Volume: COVID vs Non-COVID

Sum of Volume_Mn by Month_Num and Is_Covid_Period



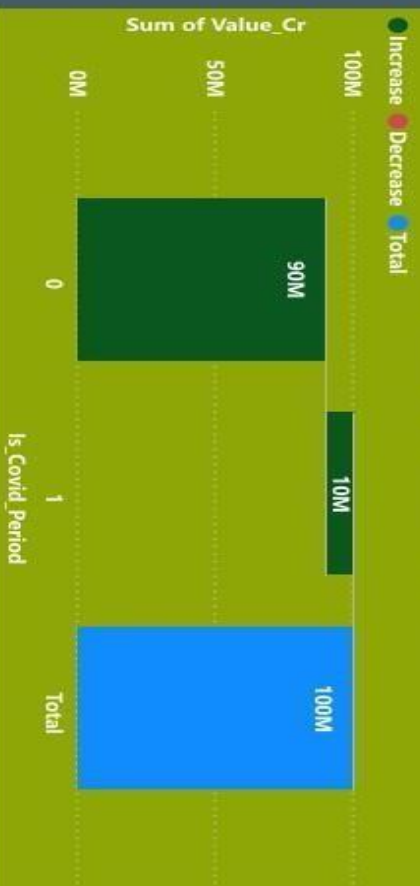
UPI Volume Share: Festive vs Non-Festive Seasons

Sum of Volume_Mn by Month_Num and Is_Festive_Season



COVID Impact on Transaction Value

Sum of Value_Cr by Is_Covid_Period



Over...

Tren...

Events

UPI Transaction Insights

AVG Transaction Value

99.99M

Average Value Growth

11.04K

AVG YoY Growth (%)

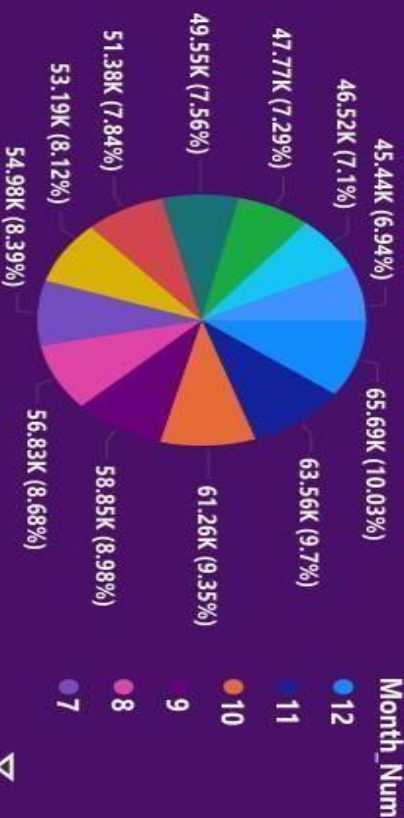
60.59K

Rolling Mean Volume

655.02K

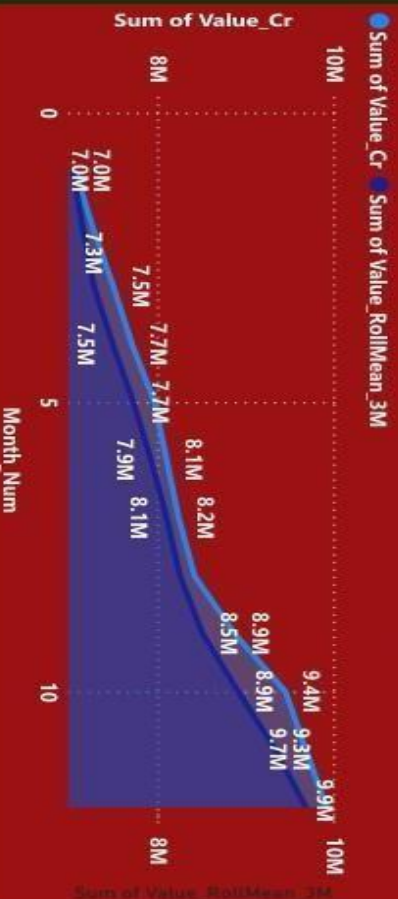
Rolling Mean Volume Trend

Sum of Volume_RollMean_3M by Month_Num



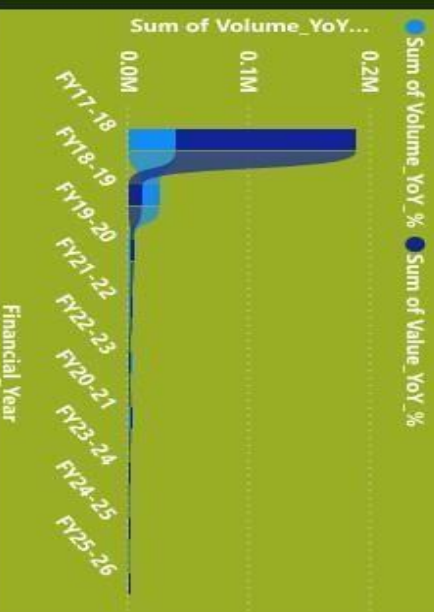
Transaction Value (Actual vs Average)

Sum of Value_Cr and Sum of Value_RollMean_3M by Month_Num



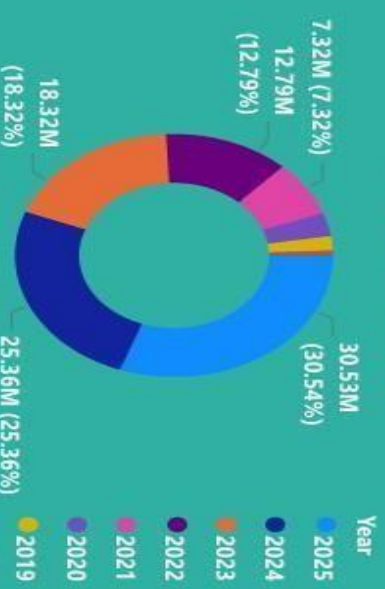
Yearly Growth Comparison (Volume vs Value %)

Sum of Volume_YoY_% and Sum of Value_YoY_% by Financial_Year



Transaction Value Distribution by Year

Sum of Value_Cr by Year



5. SOFT SKILLS DEVELOPMENT

During my internship, I not only developed technical expertise in data analytics and dashboard creation but also significantly enhanced my soft skills. These skills played a crucial role in improving my overall performance and helped me work more efficiently in a professional environment.

5.1 Communication Skills

I demonstrated strong communication skills throughout the internship by clearly expressing my ideas, asking relevant questions, and effectively interacting with mentors and peers. I was able to explain complex data insights and transaction analysis in a simple and structured manner, which helped in better understanding during discussions and review sessions. This also improved my confidence in presenting analytical findings.

5.2 Teamwork & Collaboration

I actively participated in group discussions and collaborated with peers while working on different aspects of the project. I contributed my ideas confidently and also respected others' perspectives, which helped in creating a positive and productive team environment. This experience strengthened my ability to work effectively in a team setting.

5.3 Problem-Solving Skills

While developing the Digital Payments and UPI Analysis Dashboard, I encountered challenges such as data inconsistencies, missing values, and calculation errors during data preparation and analysis. I approached these issues analytically, identified root causes, and implemented effective solutions, which improved the accuracy and

quality of the dashboard. This enhanced my analytical thinking and decision-making abilities.

5.4 Time Management

I effectively managed my time by completing assigned tasks within deadlines and maintaining consistency throughout the internship. I followed a structured approach using the log sheet, which helped me stay organized and track my daily progress. This ensured timely completion of all project activities.

5.5 Presentation Skills

I demonstrated strong presentation skills by confidently presenting my dashboard and explaining key insights during review sessions. I was able to communicate findings clearly, maintain confidence, and engage the audience effectively. My ability to simplify technical concepts into understandable insights was one of my key strengths during the internship.

5.6 Professional Growth and Learning

This internship helped me build confidence, adaptability, and a professional mindset. I improved my ability to communicate effectively, present ideas clearly, and work in a structured manner. Overall, it contributed significantly to my personal and professional development and prepared me for future career opportunities.

6. CONCLUSION

During the internship, my primary focus was on designing and developing a comprehensive Digital Payments and UPI Analysis Dashboard using data analytics and visualization techniques. This involved analyzing transaction data, identifying key performance indicators, and creating an interactive solution to effectively monitor and evaluate digital payment trends and transaction performance.

The project included multiple stages such as data collection, data cleaning, and transformation using Power BI. I created calculated measures using DAX and designed interactive dashboards with various visualizations. The key dashboards developed include transaction overview analysis, yearly and monthly trend analysis, event-based analysis, growth comparison, and transaction insights, each providing valuable information about different aspects of digital payment behavior.

Key Insights and Technical Learning

Through this internship, I significantly enhanced my knowledge in data analytics and visualization. I gained hands-on experience in Power BI, data modeling, and DAX calculations. I learned how to create interactive dashboards, calculate important performance metrics such as transaction growth percentages, rolling mean analysis, and trend indicators, and present data in a structured and meaningful way for better decision-making. This experience strengthened my understanding of business intelligence concepts and practical data analysis techniques.

Challenges Faced

While working on the project, I encountered challenges such as handling missing data, maintaining data consistency, and ensuring accuracy in calculations. Selecting appropriate visualizations to represent transaction trends and growth patterns also

required careful analysis and experimentation. Overcoming these challenges helped me improve my analytical thinking and problem-solving skills.

Real-World Application

The Digital Payments and UPI Analysis Dashboard have practical significance in the financial and digital payment sector. It helps analyze transaction behavior, monitor payment growth trends, and understand changes in transaction activity across different periods. The dashboard provides meaningful insights that can support better decision-making and help in understanding the adoption of digital payment systems.

Learning Outcome

This internship provided complete exposure to the data analytics process, from data preparation to visualization and insight generation. In addition to technical skills, I also improved my soft skills, particularly in communication, teamwork, and presentation. I developed the ability to explain analytical insights clearly, actively participate in discussions, and confidently present my work during review sessions.

Overall, the internship served as a valuable learning experience that strengthened both my technical expertise and professional skills. It enhanced my analytical thinking, problem-solving abilities, and confidence, preparing me for future opportunities in data analytics and business intelligence.

7. BIBLIOGRAPHY

The following references were used during the development of the Digital Payments and UPI Analysis Dashboard project:

- **Microsoft Power BI Documentation:**

<https://learn.microsoft.com/en-us/power-bi/>

- **DAX Reference (Microsoft):**

<https://learn.microsoft.com/en-us/dax/>

- **Microsoft Excel Support:**

<https://support.microsoft.com/excel>

- **W3Schools (Data & Web Concepts):**

<https://www.w3schools.com/>

- **Google Scholar (Research References):**

<https://scholar.google.com/>

- **YouTube (Learning Tutorials):**

<https://www.youtube.com/>

- **GitHub (Project References & Examples):**

<https://github.com/>

LOG SHEET

SI. No	Date	Concept Learnt	Duration (in Hours)
1	23/02/2026	Excel Basics: Introduction C Interface	2 Hours
2	24/02/2026	Excel Formatting C Basic Formulas	2 Hours
3	25/02/2026	IF, COUNT, SUMIF	2 Hours
4	26/02/2026	Sorting, Filtering, Charts	2 Hours
5	27/02/2026	VLOOKUP C HLOOKUP	2 Hours
6	28/02/2026	INDEX C MATCH	2 Hours
7	03/03/2026	Nested IF C Logical Functions	2 Hours
8	04/03/2026	SUMIFS / COUNTIFS	2 Hours
9	05/03/2026	Pivot Tables	2 Hours
10	06/03/2026	Advanced Pivot C Slicers	2 Hours
11	09/03/2026	Data Validation	3 Hours
12	10/03/2026	PowerQuery Basics	3 Hours
13	11/03/2026	Excel Dashboard Creation	3 Hours
14	12/03/2026	Excel Mini Project	3 Hours
15	13/03/2026	PowerBI Introduction	3 Hours
16	16/03/2026	Data Loading C Transformation	3 Hours
17	17/03/2026	Data Modeling	3 Hours
18	18/03/2026	Basic Visualizations	3 Hours
19	19/03/2026	Advanced Visualizations	3 Hours

20	20/03/2026	DAX Basics	3 Hours
21	21/03/2026	KPI Creation (CTR, Conversion Rate)	3 Hours
22	23/03/2026	Campaign Performance Dashboard	3 Hours
23	24/03/2026	User Behavior Dashboard	3 Hours
24	25/03/2026	Audience Insights Dashboard	3 Hours
25	26/03/2026	Engagement Quality Dashboard	3 Hours
26	29/03/2026	Meta Ads Performance Dashboard	4 Hours
27	30/03/2026	Slicers C Filters	3 Hours
28	31/03/2026	Dashboard Design C Formatting	3 Hours
29	01/04/2026	Drill-through C Interactions Report Navigation	3 Hours
30	02/04/2026	Python Basics: Variables C Data Types	3 Hours
31	03/04/2026	Conditional Statements C Loops	3 Hours
32	04/04/2026	Functions C Lists	3 Hours
33	05/04/2026	Dictionaries C Tuples	3 Hours
34	06/04/2026	NumPy Basics	3 Hours
35	7/04/2026	Pandas Introduction	3 Hours
36	8/04/2026	Data Frames C Series	3 Hours
37	9/04/2026	Data Cleaning C Missing Values	3 Hours
38	10/04/2026	Exploratory Data Analysis	3 Hours
39	11/04/2026	Group By C Aggregations	3 Hours
40	12/04/2026	Pivot Tables in Pandas	3 Hours

41	13/04/2026	Matplotlib Basics	3 Hours
42	14/04/2026	Advanced Plots (Bar, Line, Pie)	3 Hours
43	15/04/2026	SQL Basics: SELECT, WHERE	3 Hours
44	16/04/2026	GROUPBY, HAVING	3 Hours
45	17/04/2026	Aggregate Functions	3 Hours
46	18/04/2026	SQL JOINS	3 Hours
47	19/04/2026	Data Integration (Excel+ SQL+Power BI)	3 Hours
48	20/04/2026	Dashboard Optimization	3 Hours
49	21/04/2026	Final Dashboard Improvements	3 Hours
50	22/04/2026	Report Documentation	3 Hours
51	23/04/2026	Final Project Presentation	3 Hours
52	24/04/2026	Internship Completion Summary	3 Hours

Total Duration: 150 hours

SOFT SKILLS

1	24/02/2026	Communication Skills	2 Hours
2	25/02/2026	Active Listening	2 Hours
3	27/02/2026	Verbal Communication	2 Hours
4	03/03/2026	Teamwork	3 Hours
5	05/03/2026	Confidence Building	2 Hours

6	12/03/2026	Time Management	2 Hours
7	16/03/2026	Problem-Solving	2 Hours
8	20/03/2026	Critical Thinking	2 Hours
9	21/03/2026	Collaboration	2 Hours
10	23/03/2026	Adaptability	2 Hours
11	5/04/2026	Presentation Skills	3 Hours
12	12/04/2026	Professional Communication	3 Hours
13	15/04/2026	Responsibility	2 Hours
14	16/04/2026	Professional Growth	3 Hours
15	20/04/2026	Resume Building	3 Hours
16	21/04/2026	Interview Skills	2 Hours
17	22/04/2026	Workplace Etiquette	1 Hours
18	23/04/2026	Mock interview	2 Hours
Total Duration: 40 Hours			

Declaration: It is hereby declared that the student has completed his internship in our organization as per the above schedule.

Signature of the Authority